Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/04/2015
PI Name:	Hammond, Timothy M.D.		
Project Title:	Evolution of Genotypic and Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique to Microgravity		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
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Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 7/15/2016, per NSSC information (Ed., 4/20/15)		
Key Personnel Changes/Previous PI:	May 2015 report: Added Co-I Guri Giaever Ph.D., Associate Professor, University of British Columbia, BC, Canada. E-mail: g.giaever@ubc.ca ; Telephone (604) 827-1789		
COI Name (Institution):	Birdsall, Holly (Baylor College of Medicine) Nislow, Corey (Stanford University/University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)		
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Task Description:

This application investigates microbial growth and physiological responses to the multiple stimuli encountered in space flight environments, using Saccharomyces cerevisiae as a model system. Yeast are an ideal model because their complete gene sequence is known; there are extensive and sophisticated clone libraries available for study; and they can easily be flown in space, held at +4oC in a quiescent state, and activated by simple warming. Regulatory mechanisms in yeast are highly similar to mammalian cells. We have shown that microgravity activates a unique set of genes in yeast, but the specific molecular mechanisms remain incompletely defined.
In a corollary set of experiments, we are utilizing next generation sequencing techniques to assess the effects of long term exposure to radiation on the yeast model organism Saccharomyces cerevisiae. The cellular response to DNA damage has been studied extensively in a variety of model systems. The most comprehensive view of this process has come from yeast which, owing to the conservation of this essential pathway, shares homologs of their human DNA repair counterparts. A key observation from our group and others is that different types of DNA damage require unique repair genes and proteins. Accordingly, we are studying the cellular response to energetic particle radiation experienced in space using whole genome sequencing of thousands of yeast cells exposed to a large matrix of space-specific conditions.

We hypothesize that by identifying specific regulatory motifs within the promoter regions of microgravity-responsive genes, we will be able to determine transcription factors that bind the motifs, and thus the signaling pathways involved in microbial growth and physiological responses. We propose ground-based studies with transition to a future flight experiment, designed so as to have a high probability of obtaining genome wide data.

The proposed experiments are timely because (a) recent improvements in promoter analysis have vastly increased the accuracy of promoter elements predictions; (b) novel yeast clones allow direct measurement of gene expression by fluorescence using the plate reader in the Advanced Biological Research System (ABRS) that is already in place on the International Space Station (ISS); (c) yeast on agar plates can be assayed repetitively, thereby eliminating sample to sample variability to increasing the statistical power of the data; (d) new reagents allow design of experiments with minimal up mass (less than 2 kg); and (e) all data can be returned by telemetry, eliminating the need for down mass.

To test our hypothesis, we will:

1) Identify promoter motifs associated with genes responding to Rotating Wall Vessel (RWV)-emulated microgravity compared to static cultures. The RWV will be used to emulate microgravity as we have already demonstrated that it induces changes in yeast gene expression that are similar to those of yeast flown in space. The Yeast Green Fluorescent Protein (GFP) clone collection will permit identification, by changes in fluorescence, of genes whose expression changes in response to real and simulated microgravity. After extensive normalization and analytic procedures, microgravity-responsive genes will be grouped into clusters on the basis of similar kinetics, under the assumption that these are likely to be co-regulated. We will then search the upstream sequence of co-expressed genes and identify candidate binding motifs for transcription factors.

2) Confirm the role of regulatory sequences in directly activating transcription of a set of genes changed by RWV-simulated microgravity and to identify transcription factors that interact with these regulatory motifs. This specific aim makes novel use of the yeast deletion collection, a systematic gene-by-gene set of complete gene deletions, the only such resource that exists for any group of organisms. Transcription factors will be identified by robotically introducing a reporter plasmid that codes for 1) a suicide-inducing enzyme that is activated in all clones except the one(s) with a deletion in the gene for the transcription activating factor that binds to the promoter motif or 2) an endogenously activated essential gene to identify repressors.

3) Confirm and extend the results obtained in our ground based studies on the selective growth pressure in the true microgravity environment. Utilizing the genome wide yeast GFP clone collections spotted on agar plates, we propose a parallel flight study. Our question can ultimately only be definitively answered in the true microgravity of space, because on the ground gravity must always be onset with an equal and opposite force.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This works aims to understand the genetic control mechanism that controls the gene expression response to cellular responses to microgravity in yeast.
Yeast have 70% homolog with the human genome, and are a useful model system to define primary and secondary drug metabolism pathways that cannot be done in current mammalian cell culture systems.

The short term aim is to map the genetic promoter elements that mediate gene responses to microgravity. The long term goal is to use this information to unveil drug metabolism pathways in the unique metabolic environment provided by microgravity.

We propose to establish a sustained genomics platform on the ISS to understand the long term effects of the radiation, and other features of the ISS environment, on crew health by dissecting the essential processes of DNA damage and DNA repair. This has broad Earth benefits in understanding effects and optimization of medically necessary radiation therapy and chemotherapy with alkylating agents.

Task Progress:	Our laboratory has used both wild type yeast and the systematic yeast deletion collection to assign gene function and to understand the effect of over 3000 environmental stresses on how genotype influences phenotype. We have defined the organismal response to diverse genotoxic stresses, including drug-induced DNA damage as well as the effect of diverse types of radiation damage. These screens have i) identified genes in well-characterized DNA-damage-response pathways and ii) genes whose role in the DNA-damage response had not been previously established. Each DNA damaging treatment produced a unique genome-wide profile or "signature" and bioinformatic clustering of the data allowed us to define the genetic determinants required for diverse types of damage. In the first phase of this project (SpaceX4 and SpaceX6) we used 100 independent yeast strains to survey the effects of radiation on this model organism, and collected whole genome sequence data for 1000 genomes on space-based and ground controls. This cohort comprises 100 individual strains, 90 of which contain verified deletion alleles in a variety of conserved pathways including MEC1, MRE11, and RAD4, 5, 7, 18, 24, 27, 27, and RAD51 mutants, encoding essential proteins including checkpoint kinases, helicases, nucleases, and ligases. These 1000 genomes provide a baseline spectrum of mutations against which we will calibrate additional experimental samples. As a testament to the robustness of this platform, a replicate set of samples originally intended to return on SpaceX4 after 30 days of exposure actually remained on the ISS for 180 days, returning on SpaceX6. This allowed us to double the size of our dataset and provided insight into the effects of long-term exposure. The yeast libraries have been constructed and sequencing is ongoing. Early results are available for analysis and being studied. We are dissecting the essential processes of DNA damage and DNA repair. Whole Genome Sequencing (WGS) of thousands of strains collected over hundreds of generations should uncover space-specific vulnerabilities, and pathway analysis will allow analysis of known and novel radiation sensitive pathways. The data sets will be publicly assessable on an open-source, community database, as soon as current analysis is complete.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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